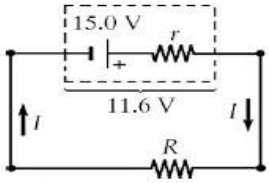
1. A battery has an emf of 15.0 V. The terminal voltage **M** of the battery is 11.6 V when it is delivering 20.0 W of power to an external load resistor R. (a) What is the value of R? (b) What is the internal resistance of the battery?

(a) Combining Joule's law, $P = I\Delta V$, and the definition of resistance, $\Delta V = IR$, gives

$$R = \frac{(\Delta V)^2}{P} = \frac{(11.6\ \text{V})^2}{20.0\ \text{W}} = \boxed{6.73\ \Omega}$$



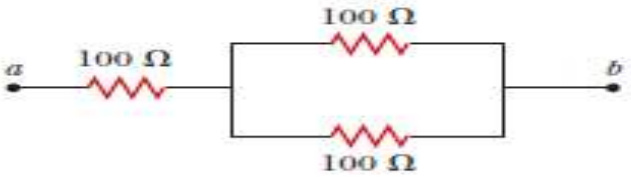
ANS. FIG. P28.1

(b) The electromotive force of the battery must equal the voltage drops across the resistances: $\mathcal{E} = IR + Ir$, where $I = \Delta V / R$.

$$r = \frac{(\mathcal{E} - IR)}{I} = \frac{(\mathcal{E} - \Delta V)R}{\Delta V}$$

$$= \frac{(15.0\ \text{V} - 11.6\ \text{V})(6.73\ \Omega)}{11.6\ \text{V}} = \boxed{1.97\ \Omega}$$

5. Three 100-Ω resistors are connected as shown in Figure P28.5. **W** The maximum power that can safely be delivered to any one resistor is 25.0 W. (a) What is the maximum potential difference that can be applied to the terminals a and b? (b) For the voltage determined in part (a), what is the power delivered to each resistor? (c) What is the total power delivered to the combination of resistors?



(a) Since all the current in the circuit must pass through the series 100-Ω resistor,

$$P_{max} = I_{max}^2 R$$

so $$I_{max} = \sqrt{\frac{P}{R}} = \sqrt{\frac{25.0\ \text{W}}{100\ \Omega}} = 0.500\ \text{A}.$$

$$R_{eq} = 100\ \Omega + \left(\frac{1}{100} + \frac{1}{100}\right)^{-1} \Omega = 150\ \Omega$$

(b) From *a* to *b* in the circuit, the power delivered is

$$P_{series} = \boxed{25.0\ \text{W}}$$ for the first resistor, and

$$P_{parallel} = I^2R = (0.250\ \text{A})^2(100\ \Omega) = \boxed{6.25\ \text{W}}$$

for each of the two parallel resistors.

(c) $P = I\Delta V = (0.500\ \text{A})(75.0\ \text{V}) = \boxed{37.5\ \text{W}}$

7. What is the equivalent resistance of the combination of identical resistors between points *a* and *b* in Figure P28.7?

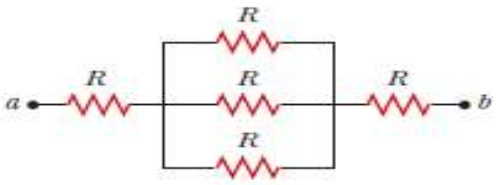


Figure P28.7

The equivalent resistance of the parallel combination of three identical resistors is

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{3}{R} \quad \text{or} \quad R_p = \frac{R}{3}$$

The total resistance of the series combination between points *a* and *b* is then

$$R_{ab} = R + R_p + R = 2R + \frac{R}{3} = \boxed{\frac{7}{3}R}$$

13. M (a) Find the equivalent resistance between points *a* and *b* in Figure P28.13. (b) Calculate the current in each resistor if a potential difference of 34.0 V is applied between points *a* and *b*.

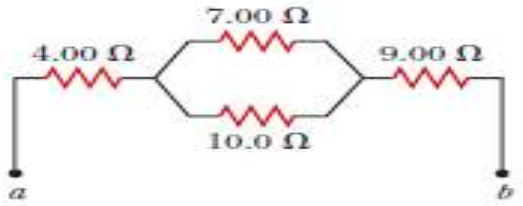
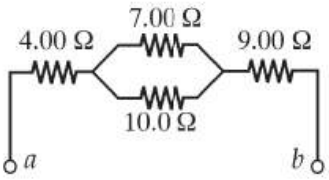


Figure P28.13

(a) The equivalent resistance of the two parallel resistors is

$$R_p = \frac{1}{(1/7.00\ \Omega) + (1/10.0\ \Omega)} = 4.12\ \Omega$$



ANS. FIG. P28.13

Thus,

$$R_s = R_1 + R_2 + R_3 = 4.00 + 4.12 + 9.00$$

$$= \boxed{17.1\ \Omega}$$

(b) $\Delta V = IR$

$34.0\ V = I(17.1\ \Omega)$

$I = \boxed{1.99\ A}$ for the 4.00-Ω and 9.00-Ω resistors.

Applying $\Delta V = IR$,

$(1.99\ A)(4.12\ \Omega) = 8.18\ V$

$8.18\ V = I(7.00\ \Omega)$

so $I = \boxed{1.17\ A}$ for the 7.00-Ω resistor. Finally,

$8.18\ V = I(10.0\ \Omega)$

so $I = \boxed{0.818\ A}$ for the 10.0-Ω resistor.

17. Consider the combination of resistors shown in Figure P28.17. (a) Find the equivalent resistance between points *a* and *b*. (b) If a voltage of 35.0 V is applied between points *a* and *b*, find the current in each resistor.

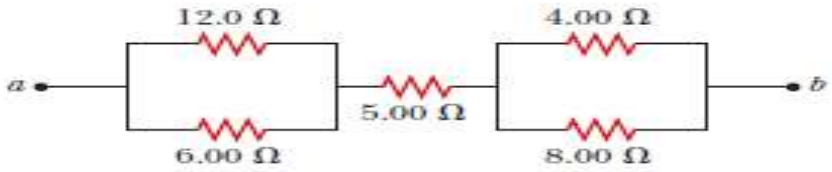


Figure P28.17

(a) The parallel combination of the 6.0 Ω and 12 Ω resistors has an equivalent resistance of

$$\frac{1}{R_{p1}} = \frac{1}{6.0\ \Omega} + \frac{1}{12\ \Omega} = \frac{2+1}{12\ \Omega}$$

or $$R_{p1} = \frac{12\ \Omega}{3} = 4.0\ \Omega$$

Similarly, the equivalent resistance of the 4.0 Ω and 8.0 Ω parallel combination is

$$\frac{1}{R_{p2}} = \frac{1}{4.0\ \Omega} + \frac{1}{8.0\ \Omega} = \frac{2+1}{8.0\ \Omega}$$

or $$R_{p2} = \frac{8\ \Omega}{3}$$

The total resistance of the series combination between points *a* and *b* is then

$$R_{ab} = R_{p1} + 5.0\ \Omega + R_{p2} = 4.0\ \Omega + 5.0\ \Omega + \frac{8.0}{3}\ \Omega$$

$$= \frac{35}{3}\ \Omega = \boxed{11.7\ \Omega}$$

(b) If $\Delta V_{ab} = 35$ V, the total current from *a* to *b* is $I_{ab} = \Delta V_{ab}/R_{ab} = 35\text{ V}/(35\ \Omega/3) = 3.0$ A and the potential differences across the two parallel combinations are

$$\Delta V_{p1} = I_{ab}R_{p1} = (3.0\text{ A})(4.0\ \Omega) = 12\text{ V}$$

and $$\Delta V_{p2} = I_{ab}R_{p2} = (3.0\text{ A})\left(\frac{8.0}{3}\ \Omega\right) = 8.0\text{ V}$$

so the individual currents through the various resistors are:

$$I_{12} = \Delta V_{p1}/12\ \Omega = \boxed{1.0\text{ A}}$$

$$I_6 = \Delta V_{p1}/6.0\ \Omega = \boxed{2.0\text{ A}}$$

$$I_5 = I_{ab} = \boxed{3.0\text{ A}}$$

$$I_8 = \Delta V_{p2}/8.0\ \Omega = \boxed{1.0\text{ A}}$$

and $$I_4 = \Delta V_{p2}/4.0\ \Omega = \boxed{2.0\text{ A}}$$

19. **W** Calculate the power delivered to each resistor in the circuit shown in Figure P28.19.

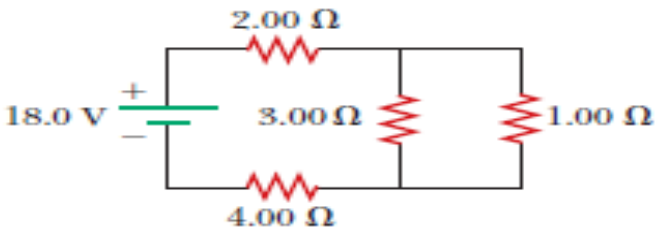
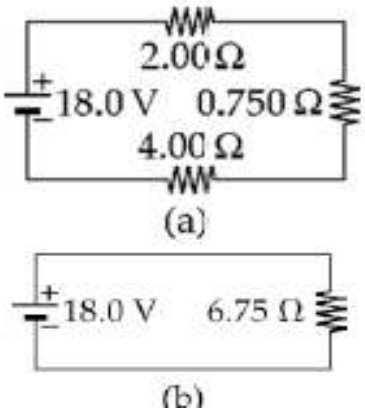


Figure P28.19

To find the current in each resistor, we find the resistance seen by the battery. The given circuit reduces as shown in ANS. FIG. P28.19(a), since

$$\frac{1}{(1/1.00\ \Omega) + (1/3.00\ \Omega)} = 0.750\ \Omega$$



In ANS. FIG. P28.19(b),

$$I = 18.0\ \text{V}/6.75\ \Omega = 2.67\ \text{A}$$

This is also the current in ANS. FIG. P28.19(a), so the 2.00-Ω and 4.00-Ω resistors convert powers

$$P_2 = I\Delta V = I^2R = (2.67\ \text{A})^2(2.00\ \Omega) = \boxed{14.2\ \text{W}}$$

and $P_4 = I^2R = (2.67\ \text{A})^2(4.00\ \Omega) = \boxed{28.4\ \text{W}}$

The voltage across the 0.750-Ω resistor in ANS. FIG. P28.19(a), and across both the 3.00-Ω and the 1.00-Ω resistors in Figure P28.19, is

$$\Delta V = IR = (2.67\ \text{A})(0.750\ \Omega) = \boxed{2.00\ \text{V}}$$

Then for the 3.00-Ω resistor,

$$I = \frac{\Delta V}{R} = \frac{2.00\ \text{V}}{3.00\ \Omega}$$

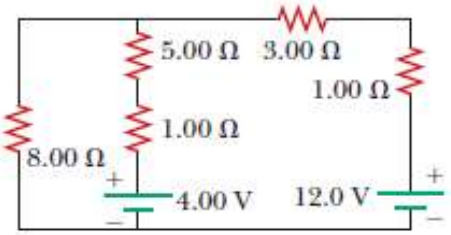
and the power is

$$P_3 = I\Delta V = \left(\frac{2.00\ \text{V}}{3.00\ \Omega}\right)(2.00\ \text{V}) = \boxed{1.33\ \text{W}}$$

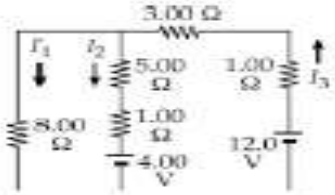
For the 1.00-Ω resistor,

$$I = \frac{2.00\ \text{V}}{1.00\ \Omega} \quad \text{and} \quad P_1 = \left(\frac{2.00\ \text{V}}{1.00\ \Omega}\right)(2.00\ \text{V}) = \boxed{4.00\ \text{W}}$$

23. **M** The circuit shown in Figure P28.22 is connected for 2.00 min. (a) Determine the current in each branch of the circuit. (b) Find the energy delivered by each battery. (c) Find the energy delivered to each resistor. (d) Identify the type of energy storage transformation that occurs in the operation of the circuit. (e) Find the total amount of energy transformed into internal energy in the resistors.



We name currents I_1, I_2, and I_3 as shown in ANS. FIG. P28.23. From Kirchhoff's current rule, $I_3 = I_1 + I_2$.



ANS. FIG. P28.23

Applying Kirchhoff's voltage rule to the loop containing I_2 and I_3,

$$12.0\ \text{V} - (4.00\ \Omega)I_3 - (6.00\ \Omega)I_2 - 4.00\ \text{V} = 0$$

$$8.00 = (4.00)I_3 + (6.00)I_2$$

Applying Kirchhoff's voltage rule to the loop containing I_1 and I_2,

$$-(6.00\ \Omega)I_2 - 4.00\ \text{V} + (8.00\ \Omega)I_1 = 0$$

or

$$(8.00\ \Omega)I_1 = 4.00 + (6.00\ \Omega)I_2$$

Solving the above linear system (by substituting $I_1 + I_2$ for I_3), we proceed to the pair of simultaneous equations:

$$\begin{cases} 8 = 4I_1 + 4I_2 + 6I_2 \\ 8I_1 = 4 + 6I_2 \end{cases} \quad \text{or} \quad \begin{cases} 8 = 4I_1 + 10I_2 \\ I_2 = \frac{4}{3}I_1 - \frac{2}{3} \end{cases}$$

and to the single equation

$$8 = 4I_1 + 10\left(\frac{4}{3}I_1 - \frac{2}{3}\right) = \frac{52}{3}I_1 - \frac{20}{3}$$

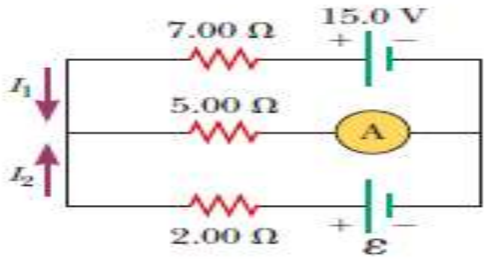
which gives

$$I_1 = \frac{3}{52}\left(8 + \frac{20}{3}\right) = 0.846\ \text{A}$$

Then

$$I_2 = I_2 = \frac{4}{3}(0.846) - \frac{2}{3} = 0.462$$

29. The ammeter shown in Figure P28.29 reads 2.00 A. **W** Find (a) I_1, (b) I_2, and (c) $\mathcal{E}$.



(a) For the upper loop:

$$+15.0\ \text{V} - (7.00\ \Omega)I_1 - (2.00\ \text{A})(5.00\ \Omega) = 0$$

$$5.00 = 7.00I_1 \quad \text{so} \quad \boxed{I_1 = 0.714\ \text{A}}$$

(b) For the center-left junction:

$$I_3 = I_1 + I_2 = 2.00\ \text{A}$$

where I_3 is the current through the ammeter (assumed to travel to the right):

$$0.714 + I_2 = 2.00 \quad \text{so} \quad \boxed{I_2 = 1.29\ \text{A}}$$

(c) For the lower loop:

$$+\mathcal{E} - (2.00\ \Omega)(1.29\ \text{A}) - (5.00\ \Omega)(2.00\ \text{A}) = 0 \rightarrow \boxed{\mathcal{E} = 12.6\ \text{V}}$$

27. Taking $R = 1.00\ \text{k}\Omega$ and $\mathcal{E} = 250\ \text{V}$ in Figure P28.27, determine the direction and magnitude of the current in the horizontal wire between a and e.

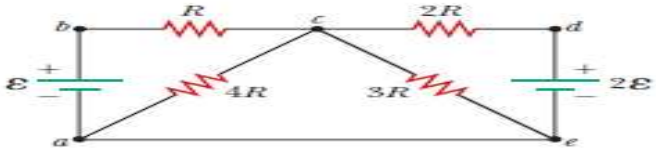
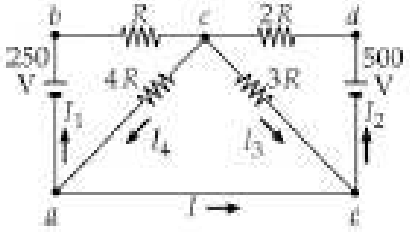
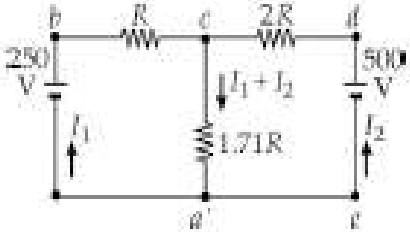


Figure P28.27



ANS. FIG. P28.27(a)



Label the currents in the branches as shown in ANS. FIG. P28.27(a). Reduce the circuit by combining the two parallel resistors as shown in ANS. FIG. P28.27(b).

Apply Kirchhoff's loop rule to both loops in ANS. FIG. P28.27(b) to obtain:

$$(2.71R)I_1 + (1.71R)I_2 = 250\ \text{V}$$

$$(1.71R)I_1 + (3.71R)I_2 = 500\ \text{V}$$

With $R = 1\,000\ \Omega$, simultaneous solution of these equations yields:

$$I_1 = 10.0\ \text{mA}$$

$$I_2 = 130.0\ \text{mA}$$

From ANS. FIG. P28.27(b), $V_c - V_a = (I_1 + I_2)(1.71R) = 240\ \text{V}$.

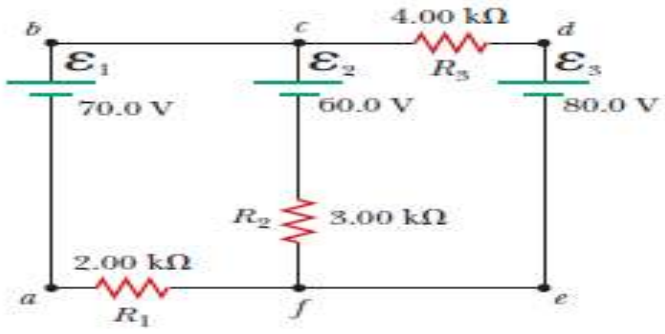
Thus, from ANS. FIG. P28.27(a), $I_4 = \dfrac{V_c - V_a}{4R} = \dfrac{240\ \text{V}}{4\,000\ \Omega} = 60.0\ \text{mA}$.

Finally, applying Kirchhoff's point rule at point a in ANS. FIG. P28.27(a) gives:

$$I = I_4 - I_1 = 60.0\ \text{mA} - 10.0\ \text{mA} = +50.0\ \text{mA}$$

or $I =$ **50.0 mA from point a to point e**.

31. M Using Kirchhoff's rules, (a) find the current in each resistor shown in Figure P28.31 and (b) find the potential difference between points *c* and *f*.



(a) We name the currents I_1, I_2, and I_3 as shown in ANS. FIG. P28.31. Applying Kirchhoff's loop rule to loop *abcfa* gives

$$+\mathcal{E}_1 - \mathcal{E}_2 - R_2 I_2 - R_1 I_1 = 0$$

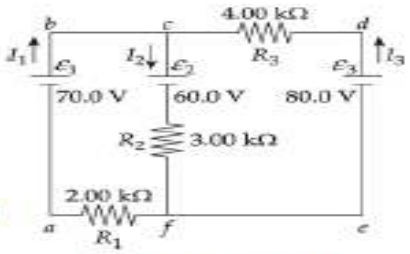
or,

$$70.0\ \text{V} - 60.0\ \text{V} - (3.00\ \text{k}\Omega)\, I_2 - (2.00\ \text{k}\Omega)\, I_1 = 0$$

which gives

$$3I_2 + 2I_1 = 10.0\ \text{mA}$$

or $$I_1 = 5.00\ \text{mA} - 1.50 I_2 \qquad [1]$$



ANS. FIG. P28.31

Applying the loop rule to loop *adcfa* yields

$$+\mathcal{E}_3 - R_3 I_3 - \mathcal{E}_2 - R_2 I_2 = 0$$

which gives

$$80.0\ \text{V} - (4.00\ \text{k}\Omega)\, I_3 - 60.0\ \text{V} - (3.00\ \text{k}\Omega)\, I_2 = 0$$

or $$3I_2 + 4I_3 = 20.0\ \text{mA}$$

and $$I_3 = 5.00\ \text{mA} - 0.750 I_2 \qquad [2]$$

Finally, applying Kirchhoff's junction rule at junction *c* gives

$$I_2 = I_1 + I_3 \qquad [3]$$

Substituting equations [1] and [2] into [3] yields

$$I_2 = 5.00\ \text{mA} - 1.50 I_2 + 5.00\ \text{mA} - 0.750 I_2$$

or $$3.25 I_2 = 10.0\ \text{mA}$$

This gives $I_2 = \dfrac{10.0\text{ mA}}{3.25} = \boxed{3.08\text{ mA}}$. Then, equation [1] yields

$$I_1 = 5.00\text{ mA} - 1.50I_2 = 5.00\text{ mA} - 1.50(3.08\text{ mA}) = \boxed{0.385\text{ mA}}$$

and from equation [2],

$$I_3 = 5.00\text{ mA} - 0.750I_2 = 5.00\text{ mA} - 0.750(3.08\text{ mA})$$
$$= \boxed{2.69\text{ mA}}$$

(b) Start at point *c* and go to point *f*, recording changes in potential to obtain

$$V_f - V_c = -\mathcal{E}_2 - R_2I_2$$
$$= -60.0\text{ V} - \left(3.00\times10^3\ \Omega\right)\left(3.08\times10^{-3}\text{ A}\right) = -69.2\text{ V}$$

or $|\Delta V|_{cf} = \boxed{69.2\text{ V and point c is at the higher potential}}$.

37. An uncharged capacitor and a resistor are connected in series to a source of emf. If $\mathcal{E} = 9.00$ V, $C = 20.0\ \mu$F, and $R = 100\ \Omega$, find (a) the time constant of the circuit, (b) the maximum charge on the capacitor, and (c) the charge on the capacitor at a time equal to one time constant after the battery is connected.

(a) The time constant of the circuit is

$$\tau = RC = (100\ \Omega)\left(20.0\times10^{-6}\text{ F}\right) = 2.00\times10^{-3}\text{ s} = \boxed{2.00\text{ ms}}$$

(b) The maximum charge on the capacitor is given by Equation 28.13:

$$Q_{max} = C\mathcal{E} = \left(20.0\times10^{-6}\text{ F}\right)(9.00\text{ V}) = \boxed{1.80\times10^{-4}\text{ C}}$$

(c) We use $q(t) = Q_{max}\left(1 - e^{-t/RC}\right)$, when $t = RC$. Then,

$$q(t) = Q_{max}\left(1 - e^{-RC/RC}\right) = Q_{max}\left(1 - e^{-1}\right) = \left(1.80\times10^{-4}\text{ C}\right)\left(1 - e^{-1}\right)$$
$$= \boxed{1.14\times10^{-4}\text{ C}}$$

38. W Consider a series RC circuit as in Figure P28.38 for which $R = 1.00\ \text{M}\Omega$, $C = 5.00\ \mu\text{F}$, and $\mathcal{E} = 30.0\ \text{V}$. Find (a) the time constant of the circuit and (b) the maximum charge on the capacitor after the switch is thrown closed. (c) Find the current in the resistor 10.0 s after the switch is closed.

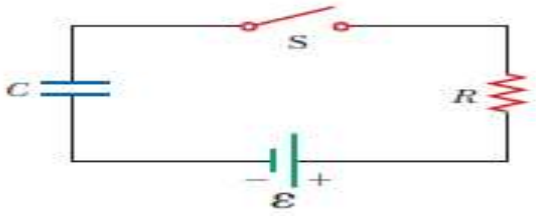


Figure P28.38

(a) The time constant is

$$RC = (1.00\times10^{6}\ \Omega)(5.00\times10^{-6}\ \text{F}) = \boxed{5.00\ \text{s}}$$

(b) After a long time interval, the capacitor is "charged to thirty volts," separating charges of

$$Q = C\mathcal{E} = (5.00\times10^{-6}\ \text{C})(30.0\ \text{V}) = \boxed{150\ \mu\text{C}}$$

(c) $$I(t) = \frac{\mathcal{E}}{R}e^{-t/RC} = \left(\frac{30.0\ \text{V}}{1.00\times10^{6}\ \Omega}\right)\exp\left[\frac{-10.0\ \text{s}}{(1.00\times10^{6}\ \Omega)(5.00\times10^{-6}\ \text{F})}\right]$$

$$= \boxed{4.06\ \mu\text{A}}$$

39. W A 2.00-nF capacitor with an initial charge of 5.10 μC is discharged through a 1.30-kΩ resistor. (a) Calculate the current in the resistor 9.00 μs after the resistor is connected across the terminals of the capacitor. (b) What charge remains on the capacitor after 8.00 μs? (c) What is the maximum current in the resistor?

(a) From $I(t) = -I_0 e^{-t/RC}$,

$$I_0 = \frac{Q}{RC} = \frac{5.10\times10^{-6}\ \text{C}}{(1\,300\ \Omega)(2.00\times10^{-9}\ \text{F})} = 1.96\ \text{A}$$

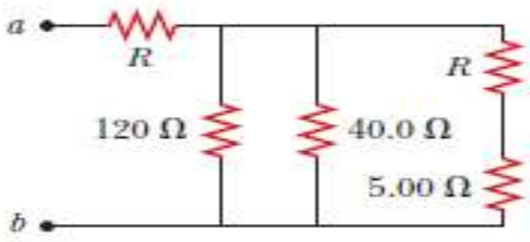
$$I(t) = -(1.96\ \text{A})\exp\left[\frac{-9.00\times10^{-6}\ \text{s}}{(1\,300\ \Omega)(2.00\times10^{-9}\ \text{F})}\right] = \boxed{-61.6\ \text{mA}}$$

(b) $$q(t) = Qe^{-t/RC} = (5.10\ \mu\text{C})\exp\left[\frac{-8.00\times10^{-6}\ \text{s}}{(1\,300\ \Omega)(2.00\times10^{-9}\ \text{F})}\right]$$

$$= \boxed{0.235\ \mu\text{C}}$$

(c) The magnitude of the maximum current is $I_0 = \boxed{1.96\ \text{A}}$.

56. The resistance between terminals a and b in Figure P28.56 is 75.0 Ω. If the resistors labeled R have the same value, determine R.



The equivalent resistance is $R_{eq} = R + R_p$, where R_p is the total resistance of the three parallel branches;

$$R_p = \left(\frac{1}{120\ \Omega} + \frac{1}{40.0\ \Omega} + \frac{1}{R + 5.00\ \Omega}\right)^{-1}$$

$$= \left(\frac{1}{30.0\ \Omega} + \frac{1}{R + 5.00\ \Omega}\right)^{-1}$$

$$= \frac{(30.0\ \Omega)(R + 5.00\ \Omega)}{R + 35.0\ \Omega}$$

Thus,

$$75.0\ \Omega = R + \frac{(30.0\ \Omega)(R + 5.00\ \Omega)}{R + 35.0\ \Omega} = \frac{R^2 + (65.0\ \Omega)R + 150\ \Omega^2}{R + 35.0\ \Omega}$$

which reduces to

$$R^2 - (10.0\ \Omega)R - 2\,475\ \Omega^2 = 0$$

or

$$(R - 55\ \Omega)(R + 45\ \Omega) = 0$$

Only the positive solution is physically acceptable, so $R = \boxed{55.0\ \Omega}$.

50. Find the equivalent resistance between points *a* and *b* in Figure P28.50.

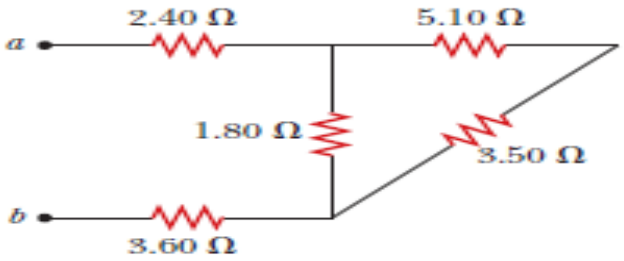
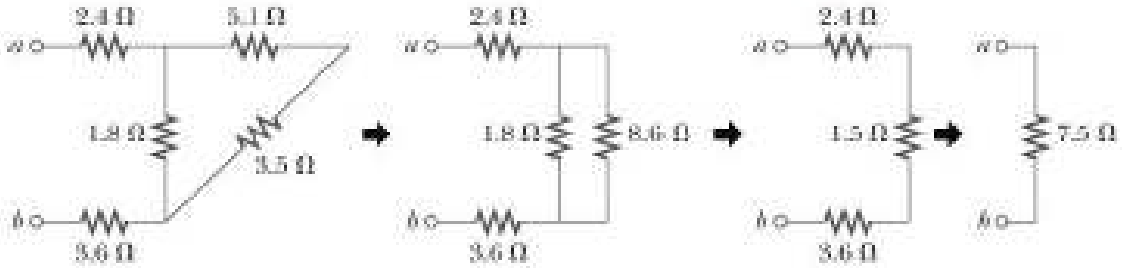


Figure P28.50

The resistive network between *a* an *b* reduces, in the stages shown in ANS. FIG. P28.50, to an equivalent resistance of $R_{eq} = \boxed{7.49\ \Omega}$.



ANS. FIG. P28.50